

Matematica - Clasa a 12-a

Exercitii - Toate capitolele

Nume: _____

Data: _____

Sectiunea 1 - Limite de functii

1. Calculeaza limitele:

$$\lim_{x \rightarrow 0} \sin(4x)/x = ______ ; \lim_{x \rightarrow 0} (e^{(2x)} - 1)/x = ______$$

$$\lim_{x \rightarrow \infty} (2x^2 + x)/(x^2 - 3) = ______ ; \lim_{x \rightarrow 1} (x^2 - 1)/(x - 1) = ______$$

$$\lim_{x \rightarrow 0} \ln(1 + 3x)/x = ______ ; \lim_{x \rightarrow 2} (x^2 - 4)/(x - 2) = ______$$

Sectiunea 2 - Derivate

2. Calculeaza derivata:

$$(x^5 - 3x^2 + 7)' = ______$$

$$(x^2 e^x)' = ______ ; (\sin^2 x)' = ______$$

$$(\ln(x^2 + 1))' = ______ ; (e^{(\sin x)})' = ______$$

$$(x/\cos x)' = ______ \text{ (cat)}$$

3. Ecuatia tangentei la $f(x) = x^3 - x$ in $x = 1$:

$$f(1) = ______ ; f'(x) = ______ ; f'(1) = ______ ; \text{Tangenta: } y = ______$$

Sectiunea 3 - Studiul functiei

4. Studiaza $f(x) = x^2 - 4x + 3$:

$$f'(x) = ______ ; f' = 0 \text{ pt } x = ______$$

$$\text{Minim/maxim local: } ______ ; \text{Valoare extrema: } ______$$

$$\text{Crescatoare pe } ______ ; \text{Descrescatoare pe } ______$$

5. Asimptotele lui $f(x) = (x^2 - 1)/x$:

$$\text{Verticala: } ______ ; \text{Oblica: } m = ______ \text{ } n = ______ \Rightarrow y = ______$$

Exercitii Clasa 12 (continuare)

Sectiunea 4 - Integrala

6. Calculeaza primitivele:

$$\int (4x^3 - 2x + 1) dx = ______$$

$$\int \sin(2x) dx = ______ ; \int e^{(3x)} dx = ______$$

$$\int x \ln x dx = ______ \text{ (prin parti)}$$

7. Integrala definita:

$$\int_{[0,2]} x^2 dx = ______ ; \int_{[0,\pi/2]} \cos x dx = ______$$

$$\int_{[1,e]} 1/x dx = ______ ; \int_{[0,1]} e^x dx = ______$$

8. Aria intre $y=x^2$ si $y=x$:

$$\text{Intersectii: } x = ______ ; A = \int_{[0,1]} (x - x^2) dx = ______$$

Sectiunea 5 - Matrice, Sisteme, Combinatorica

9. Calculeaza:

$$\det[[3,2],[1,4]] = ______ ; \det[[1,0,2],[3,1,0],[0,2,1]] = ______$$

10. Sistem Cramer: $3x - y = 5$, $x + 2y = 7 \Rightarrow x = ______, y = ______$

11. Combinatorica:

$$C(9,3) = ______ ; A(7,2) = ______ ; P(5) = ______$$

$$\text{Termenul } T_3 \text{ din } (x+2)^5 = C(5,2)x^3 \times 2^2 = ______$$

12. Probabilitati: $X \sim B(6, 0.5)$. $E(X) = ______ D(X) = ______ P(X=3) = ______$

Exercitii Clasa 12 - Probleme aplicate

Sețiunea 6 - Probleme

13. Studiu complet $f(x)=e^x(x-1)$. Extreme, inflexiune.

$f'=e^x x$, $f'=0$ pt $x=0$. $f''=e^x(x+1)=0$ pt $x=-1$ (inflexiune). Min in $x=0$.

14. Calculeaza V rotatie al lui $y=x^2$ pe $[0,1]$ in jurul Ox .

$$V=\pi \int [0,1] x^4 dx = \pi [x^5/5]_0^1 = \pi/5$$

15. Rezolva sistemul prin Gauss: $x+y+z=6$, $2x-y+z=3$, $x+2y-z=2$.

Matricea extinsa $\rightarrow$ forma treapta $\rightarrow x=---$, $y=---$, $z=---$

16. Probabilitate conditionata: boala B , test T .

$P(B)=0.01$, $P(T|B)=0.99$, $P(T|B')=0.05$. Calculeaza $P(B|T)$ (Bayes)=---

17. Suma primilor n termeni ai progresiei $1+3+9+\dots=$

$$q=3, a_1=1. S_n = (3^n - 1)/2$$

18. O functie f continua cu $f(0)=-1$ si $f(1)=2$. Exista x_0 cu $f(x_0)=0$?

Da, prin teorema valorilor intermediare (schimbare de semn)

Exercitii Clasa 12 - Exercitii suplimentare

Sectiunea 7 - Adevarat/Fals

- $\int f dx$ si $d/dx [F(x)]$ se anuleaza reciproc ---
- $\det(A+B) = \det(A) + \det(B)$ ---
- $C(n,0) = 1$ pt orice n ---
- L'Hopital se aplica la $0/0$ si $\inf/\inf$ ---
- Daca $f'(x_0) = 0 \Rightarrow x_0$ e extrem local ---
- Normala $N(0,1)$ are media 0 si dispersia 1 ---

Sectiunea 8 - Calcule rapide

19. Derivate:

$$(e^{(x^2)})' = ______ ; (\ln(\cos x))' = ______ ; (\arctan(2x))' = ______$$

20. Integrale rapide:

$$\int 6x^2 dx = ______ ; \int 1/(1+x^2) dx = ______ ; \int \sin x dx = ______$$

21. Limite rapide:

$$\lim_{x \rightarrow 0} \sin(x)/x = ______ ; \lim_{x \rightarrow \infty} 1/x^3 = ______ ; \lim_{x \rightarrow 2} (x-2)/(x^2-4) = ______$$

Sectiunea 9 - Demonstratii scurte

22. Demonstreaza ca $f(x) = x^3$ e strict crescatoare pe $\mathbb{R}$.

23. Demonstreaza identitatea: $(a+b)^2 - (a-b)^2 = 4ab$.

Exercitii Clasa 12 - Test final

24. $\lim_{x \rightarrow 0} (1 - \cos x)/x^2 = ?$ (se stie ca = $1/2$)

Raspuns: $1/2$

25. Studiu complet: $f(x) = x^2 e^{-x}$.

$f' = xe^{-x}(2-x)$. Cresc: $(0, 2)$, Desc: $(-\infty, 0)$ si $(2, +\infty)$. Max in $x=2$.

26. $\int [0, 1] x e^x dx = ?$ (parti)

$$= [xe^x - e^x]_{-0}^1 = (e - e) - (-1) = 1$$

27. Cramer: $2x + 3y = 7$, $x - y = 1$. Solutia?

$$\det = 2(-1) - 3(1) = -5. \quad x = \det(A_1)/\det(A). \quad \text{Sol: } x=2, y=1$$

28. $C(12, 3) + C(12, 4) = C(13, 4)$. Verifica!

$$C(12, 3) = 220, C(12, 4) = 495, \text{ suma} = 715 = C(13, 4) = 715 \text{ OK}$$

29. $X \sim N(5, 4)$. $P(3 < X < 7) = ?$ ($\sigma = 2$, regula 68%)

$$(3, 7) = (5 - 2, 5 + 2) = (\mu - \sigma, \mu + \sigma) \Rightarrow P \approx 68\%$$

30. Aria delimitata de $y = \sin x$ si Ox pe $[0, \pi]$.

$$A = \int [0, \pi] \sin x \, dx = [-\cos x]_{-0}^{\pi} = 1 + 1 = 2$$